

DEVADATTU KAKUMANU

Data Engineer/ Data Analyst

Location: Kansas City, MO | **Phone:** (816)-766-3454 | **Email:** saidattukakumanu@gmail.com | [LinkedIn](#)

SUMMARY

- Demonstrated over 4 years of expertise as a Data Engineer in Big Data Hadoop Ecosystem, Cloud Engineering, and Data Warehousing, managing large datasets and optimizing performance using AWS and Azure services, Hadoop, and various ETL and reporting tools.
- Leveraged AWS services (EC2, S3, EMR, RDS, VPC, Elastic Load Balancing, IAM, Auto Scaling, CloudFront, CloudWatch, Lambda) to enhance resource efficiency by 30% through automated scaling and monitoring solutions.
- Engineered data pipelines with Azure Data Factory and Azure Databricks, loading data into Azure Data Lake, Azure SQL Database, and Azure SQL Data Warehouse, improving data retrieval time by 25%.
- Optimized data processing and storage solutions using Azure services (HDInsight, Stream Analytics, Active Directory, Blob Storage, Cosmos DB), reducing costs by 20%.
- Managed datasets up to 50 TB across major Hadoop distributions (Cloudera, Amazon EMR, Azure HDInsight, Hortonworks), ensuring high availability and performance.
- Automated processes with Oozie and Airflow to handle terabytes of streaming data (Kafka, Spark streaming, Storm) and batch data, reducing processing time by 40%.
- Developed production-ready applications using Spark components (Spark SQL, Data Frames, Datasets, Spark-ML, Spark Streaming).
- Created efficient data pipelines and reduced query response time by 35% using SQL and NoSQL databases (Cosmos DB, MongoDB, HBase, Cassandra).
- Scripted with Python (PySpark), Scala, and Spark-SQL for various file formats (XML, JSON, CSV, Parquet), enhancing processing speed by 30%.
- Analyzed data using HiveQL, Hive-ACID tables, Pig Latin queries, and custom MapReduce programs, reducing data retrieval time by 20%.
- Experienced in all phases of data management, from acquisition and warehousing to modeling, processing, and transformation.
- Monitored document growth and estimated storage size for large MongoDB clusters, managing data lifecycle effectively.
- Executed Ad-hoc queries, indexing, replication, load balancing, and aggregation in MongoDB.
- Developed Kubernetes clusters using cloud formation templates and PowerShell scripting for cloud deployment automation.
- Built highly scalable and resilient Restful APIs, ETL solutions, and third-party integrations using Informatica.
- Utilized bug tracking and ticketing systems like Jira and Remedy, and version control tools such as Git and SVN.
- Managed project timelines and deliverables using Waterfall and Agile Scrum methodologies, improving on-time delivery by 15%.
- Migrated legacy applications to cloud platforms using DevOps tools (GitHub, Jenkins, JIRA, Docker, Slack), reducing downtime by 50%.
- Integrated DBT for data transformations, Snowflake for data warehousing, and Grafana for data visualization, enhancing data handling efficiency.
- Implemented ETL processes using Talend, improving data integration and transformation capabilities.

EDUCATION

University of Missouri-Kansas City (UMKC)

Master's Degree (MS) in Computer Science

Kansas, MO

Jan 2023 – May 2024

TECHNICAL SKILLS

Programming Languages:	Python, R, C, J2EE, SQL, Pig Latin, HiveQL, Scala, Unix Shell Scripting.
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, TensorFlow.
Databases:	MS-SQL SERVER, Oracle, Snowflake, MS-Access, MySQL, Teradata, PostgreSQL, DB2.
Big Data Technologies:	Yarn, MapReduce, Pig, Hive, HBase, Cassandra, Oozie, Apache Spark, Scala, Impala, Kafka.
Hadoop Distributions:	Apache Hadoop 2.x/1.x, Cloudera CDP, Hortonworks HDP, Amazon EMR (EMR, EC2, EBS, RDS, S3, Glue, Elasticsearch, Lambda, Kinesis, SQS, DynamoDB, Redshift, ECS) Azure HDInsight (Databricks, Data Lake, Blob Storage, Data Factory, SQL DB, SQL DWH, Cosmos DB, Azure DevOps, Active Directory).
NoSQL Database:	Cassandra, MongoDB.
Reporting Tools/ETL Tools:	Informatica, Talend, SSIS, SSRS, SSAS, ER Studio, DBT, Tableau, Power BI, Grafana.
Methodologies:	Agile/Scrum, Waterfall.
Development Tools:	Eclipse, NetBeans, IntelliJ, Hue, Microsoft Office Suite (Word, Excel, PowerPoint, Access)
Operating Systems:	Windows, Macintosh, Linux, Ubuntu, Unix.
Others:	Machine learning, NLP, Stream Sets, Spring Boot, Jupyter Notebook, Docker, Kubernetes, Jenkins, Jira.

PROFESSIONAL EXPERIENCE

UMKC-Sodexo, Student worker | Kansas City, MO

Aug 2023 – May 2024

- Performed data cleansing and applied transformations using Databricks and Spark data analysis, processing datasets of up to 10TB.
- Enhanced data processing efficiency by 30% utilizing Azure cloud platforms (HDInsight, Databricks, Data Lake, Blob, Data Factory, Synapse, SQL DB, SQL DWH).
- Integrated enterprise data warehousing with Big Data analytics using Azure Synapse analytics, optimizing data retrieval time by 25%.
- Developed JSON Scripts for deploying Pipelines in Azure Data Factory (ADF) to process data using SQL Activity, reducing manual workload by 40%.
- Automated workflows for daily incremental loads, transitioning data from RDBMSs to data lakes, resulting in a 35% reduction in processing time.
- Designed Pipelines in Azure Data Factory to Extract, Transform, and Load data from various sources, including Azure SQL and Blob storage.
- Structured database objects (tables, views, stored procedures, triggers, packages, functions) using T-SQL, improving query performance by 20%.

- Facilitated data for interactive Power BI dashboards and reports, increasing data visibility for stakeholders by 50%.
- Integrated data from MongoDB, MS SQL, and cloud sources (Blob, Azure SQL DB) using Azure Data Factory, SQL API, and Mongo API.
- Automated data management tasks with SQL Scripts, streamlining processes and improving efficiency by 30%.
- Orchestrated data ingestion workflows using Airflow Operators and other Python libraries, enhancing workflow automation by 25%.
- Utilized Databricks notebooks for interactive analytics with Spark APIs, improving data analysis speed by 40%.
- Engineered custom input adapters using Spark, Sqoop, and Oozie to ingest and analyze data from RDBMS to Azure Data Lake.
- Implemented solutions for SQL scripts analysis using PySpark, reducing processing time by 35%.
- Deployed Applications into Azure Cloud through Continuous Integration and Continuous Deployment (CI/CD), decreasing deployment time by 50%.
- Created interactive visualizations using PowerBI, enhancing data reporting and visualization capabilities.
- Led team communication and management, fostering collaboration, and resolving conflicts efficiently, resulting in a 20% improvement in project delivery times.

APTIV COMPONENTS, Software Engineer | Bengaluru, India

Feb 2022 – Dec 2022.

- Engineered Python-based data pipelines on Hadoop, processing datasets of 500GB+ for vehicle electrification systems, enhancing data-driven insights by 30%.
- Constructed scalable data architectures using Azure services (Data Factory, Data Lake), integrating data from 100+ sources, and improving data accessibility by 40%.
- Developed 20 interactive dashboards with Power BI and Seaborn, offering actionable insights and increasing reporting efficiency by 25%.
- Conducted statistical analysis and modeling with Python (SciPy), optimizing battery management and electric vehicle architecture, boosting performance by 15%.
- Ensured data quality and consistency through ETL processes on Hadoop, supporting high-voltage power distribution projects and maintaining 99.9% data accuracy.
- Managed 50TB+ of data on Snowflake, optimizing query performance for a 50% reduction in data retrieval times.
- Collaborated in agile environments to deliver data solutions, supporting sustainable vehicle electrification strategies, and reducing project delivery time by 20%.
- Presented technical findings in over 30 presentations to diverse stakeholders, demonstrating strong communication skills and influencing decision-making processes.
- Leveraged Spark for managing large datasets on S3 Data Lake, implementing Hive external tables, and improving data handling efficiency by 35%.
- Employed AWS services (S3, EC2, IAM, RDS) to build robust data pipelines, automating processes with Step Functions, Data Pipeline, and Glue, enhancing operational workflows by 25%.

KPMG, Data Analyst | India

Jul 2020 – Jan 2022

- Developed custom data processing scripts using Python and SQL, automating ETL tasks to enhance efficiency and accuracy in data analysis workflows.
- Employed NumPy and Pandas for data manipulation, ensuring data integrity and accuracy by cleaning, merging, and transforming datasets of up to 1 million rows.
- Integrated Matplotlib and SciPy to create insightful visualizations and statistical analysis reports, providing stakeholders with actionable insights into key business metrics and trends through 50+ visual reports.
- Utilized Tableau for data blending and creating comprehensive datasets, offering a holistic view of business metrics through 30+ interactive dashboards.
- Executed optimized SQL queries to retrieve specific datasets efficiently, enhancing database performance with indexing and query optimization techniques, leading to a 25% reduction in query response time.
- Created dynamic charts in Excel to visually represent complex data trends, aiding stakeholders in understanding key insights through 40+ detailed charts.
- Restructured Pivot tables for summarizing and analyzing large datasets, providing actionable insights for business decision-making.
- Administered AWS services like EC2 and S3 for scalable and secure data storage, facilitating efficient access and retrieval of large datasets.
- Implemented AWS Lambda functions to automate data processing tasks, reducing manual effort, and increasing operational efficiency by creating 20+ Lambda functions.
- Collaborated within Agile methodologies to prioritize and execute data analysis tasks, ensuring alignment with business objectives and timely project delivery across 300+ hours of sprints.
- Conducted thorough data cleaning and wrangling to preprocess raw data, addressing missing values and outliers to enhance analysis quality.
- Streamlined Informatica for ETL processes, creating 25+ workflows and mappings to ensure seamless data integration and transformation.
- Applied data mining techniques to uncover patterns and trends in customer behavior, product performance, and market trends, analyzing data from 10+ sources.
- Leveraged Talend for data integration, creating and managing 20+ data pipelines to streamline data processing and enhance data quality.
- Designed and executed A/B testing experiments to evaluate marketing campaign effectiveness, driving data-driven optimization efforts.
- Demonstrated strong communication skills, effectively presenting analytical findings and recommendations to stakeholders.

APTIV COMPONENTS, Software Engineer Trainee | India

Aug 2019 – Jun 2020

- Engineered ETL processes using Python, Pandas, and SQL, achieving a 40% reduction in data processing time and enhancing pipeline efficiency by 25%.
- Automated data extraction and processing with Python, Selenium, and Beautiful Soup, cutting manual effort and boosting operational efficiency.
- Developed batch and stream processing applications using Spark APIs, enhancing data throughput by 35% and streamlining functional pipelining.

- Constructed data pipelines and performed analytics with AWS services (EMR, EC2, S3, RDS, Lambda, Glue, SQS, Redshift), processing over 10TB of data monthly.
- Utilized Spark's in-memory capabilities to manage large datasets in S3 Data Lake, efficiently filtering and loading data into Hive external tables, handling datasets up to 5TB.
- Created and modified SQL stored procedures, functions, views, indexes, and triggers, optimizing database performance and query speed by 20%.
- Executed ETL operations using Python, Spark SQL, S3, and Redshift, deriving customer insights from over 15TB of data and generating 50+ detailed reports.
- Set up CI/CD pipelines using Jenkins, Terraform, and AWS, ensuring end-to-end architecture assessment and implementation of AWS services, leading to a 40% faster deployment cycle.
- Transformed data using AWS Glue dynamic frames with PySpark, cataloging transformed data with Crawlers, and scheduling jobs and crawlers using workflow features, resulting in a increase in data processing efficiency.
- Implemented continuous integration and deployment with CI tools like Jenkins and Bamboo, utilizing EC2 Container Service plugins to automate Jenkins master-slave configuration and deployment processes, reducing deployment time by 35%.

PROJECT

Used Car Price Detection

- **Objective:** The goal of this project is to simplify the complexities of the used car market by leveraging data science to establish fair and optimal prices for used vehicles. This enhances transparency and efficiency for both buyers and sellers, helping them make informed decisions.
- **Project Overview:** Buying or selling a used car involves numerous factors that influence the vehicle's fair market value. This project uses advanced machine learning algorithms to analyze extensive datasets and provide data-driven insights for accurate price estimation.

Technologies: Python, Requests, BeautifulSoup, xml, NumPy, pandas, matplotlib, seaborn, XGBoost, scikit-learn, HTML, CSS, JavaScript, Flask/Django

Meme Classification and Location Privacy Enhancement

- **Objective:** Developed a novel technique to establish location privacy based on K-Anonymity and Obfuscation, while creating an advanced system to classify memes into various emotional categories to understand their sentiment and emotional impact better.
- **Project Overview:** The project was conducted in two phases, progressively exploring, and classifying emotions in memes.

Technologies: ResNet50, VGG16, Conv2D, Dense, Dropout, Global Average Pooling, Bidirectional LSTM, Conv1D, Dense layers, NLP techniques.